

ABHISHEK S

Generative AI • Machine Learning • Software Engineering

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Professional Summary

Computer Science Engineering student focused on Artificial Intelligence, Machine Learning, and Software Development. Experienced in building Retrieval-Augmented Generation (RAG) systems, Large Language Model applications, intelligent data pipelines, and analytics platforms. Passionate about developing scalable software solutions that combine modern AI techniques with strong engineering principles to solve real-world problems.

Projects

AI Video Assistant | Generative AI, Retrieval-Augmented Generation (RAG) GitHub

- Built an AI-powered video intelligence platform capable of processing YouTube videos and audio files into searchable knowledge bases.
- Developed a Retrieval-Augmented Generation (RAG) pipeline using LangChain and ChromaDB for transcript-based question answering and semantic retrieval.
- Integrated Whisper speech recognition for automated transcription, meeting summarization, action-item extraction, and key-decision detection.
- Implemented vector embeddings, semantic search, and conversational querying across transcripts exceeding 2+ hours.
- Designed and deployed an interactive Streamlit interface supporting YouTube ingestion and local audio uploads.

Technologies: Python, LangChain, ChromaDB, Streamlit, Whisper, Transformers, Mistral AI, Vector Databases, FFmpeg, yt-dlp

SIGNAL | Deep Packet Intelligence & Threat Analytics Platform

- Architected a multi-threaded Deep Packet Inspection (DPI) engine in C++ for real-time traffic classification and network intelligence generation.
- Processed and analyzed hundreds of network flows from PCAP captures to extract protocol-level and flow-level insights.
- Developed a security analytics dashboard featuring application classification, protocol analysis, traffic visualization, and threat assessment.
- Implemented threat-scoring mechanisms leveraging DNS activity, unidentified traffic patterns, peer-to-peer communication, and abnormal transfers.
- Automated generation of PDF security reports with actionable recommendations and traffic intelligence summaries.

Technologies: C++, Python, Streamlit, Pandas, Plotly, ReportLab, Networking, PCAP Analysis, DPI

Technical Skills

Programming	Python, C++, SQL
Artificial Intelligence	Machine Learning, Generative AI, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Large Language Models (LLMs), Natural Language Processing (NLP)
Data Technologies	Pandas, NumPy, ChromaDB, Data Analysis, Data Visualization, Vector Databases, Plotly
Software Development	Git, GitHub, Linux, Streamlit, REST APIs
Networking	Computer Networks, TCP/IP, Deep Packet Inspection (DPI), Network Traffic Analysis, PCAP Analysis, Threat Analytics

Education

Marian Engineering College, Thiruvananthapuram 2023 – Present

Bachelor of Technology (B.Tech) in Computer Science and Engineering

CGPA: **9.52 / 10.00**

Bharatiya Vidya Bhavan Senior Secondary School, Kodunganoor 2023

Class XII (CBSE): **90.6%**

St. Shantal Malamukal School 2021

Class X (CBSE): **95.6%**

Relevant Coursework

Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Machine Learning, Artificial Intelligence

Achievements

- Presented cybersecurity research titled **Fortifying Digital Frontiers** at the Institution of Engineers (India) (IEI) technical event hosted at Marian Engineering College.
- Maintained a cumulative CGPA of **9.52/10.00** in Computer Science and Engineering.
- Developed projects spanning Generative AI, Retrieval-Augmented Generation, Network Analytics, Software Engineering, and Intelligent Systems.

Professional Strengths

Problem Solving • Analytical Thinking • Technical Communication • Research Presentation • Technical Documentation
• Team Collaboration • Continuous Learning